

Summary Data on Acid Cropland and Lime Requirements in Ethiopia

GAIA

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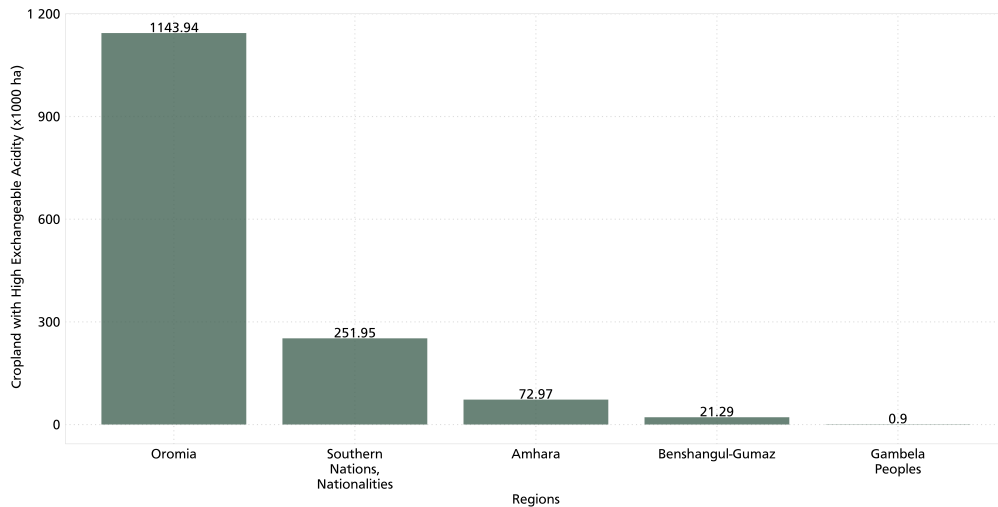
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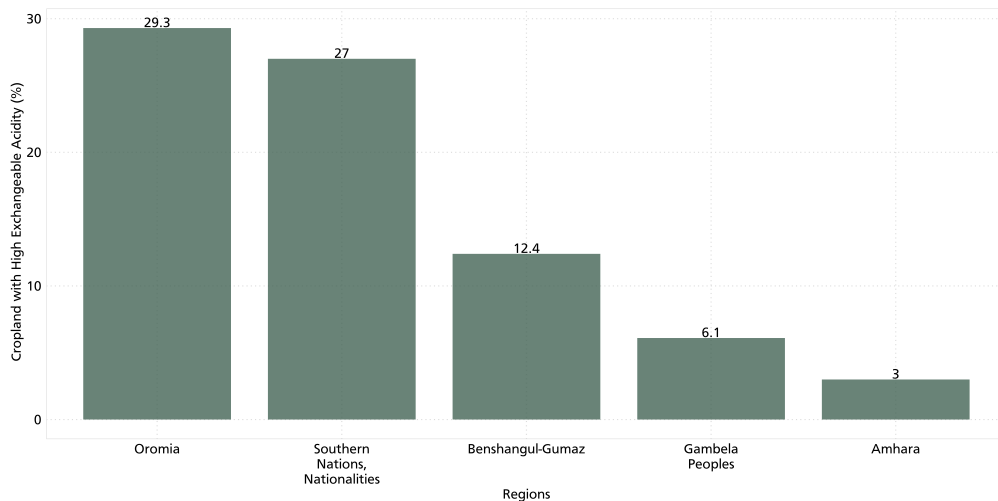
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Ethiopia**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production, ROI and profitability from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

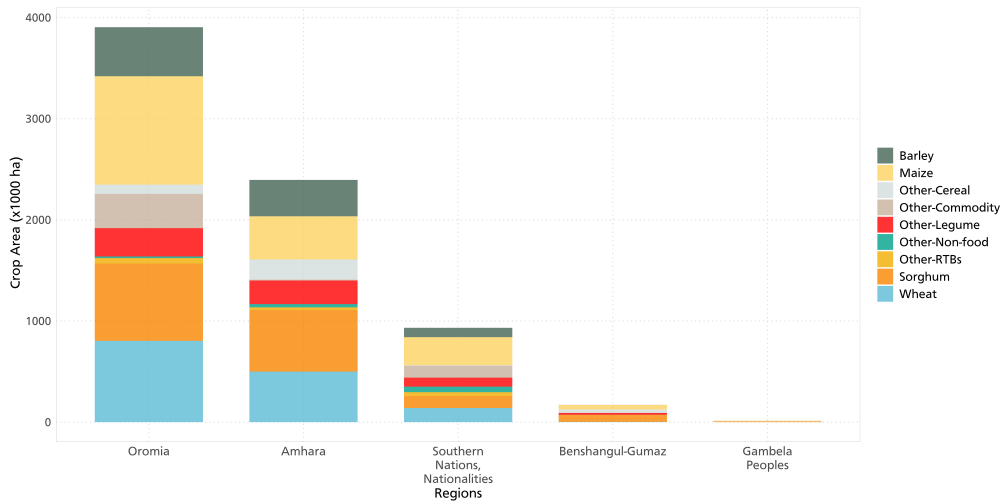
Top 10 Regions in Ethiopia with the highest acidic cropland



Top 10 Regions in Ethiopia with the highest acidic cropland (Percentage of total cropland)



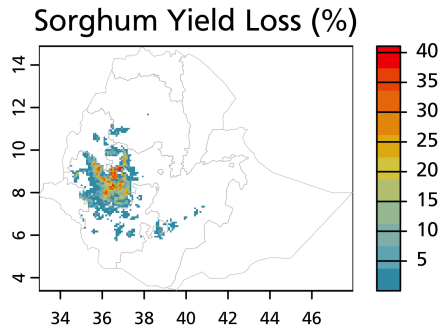
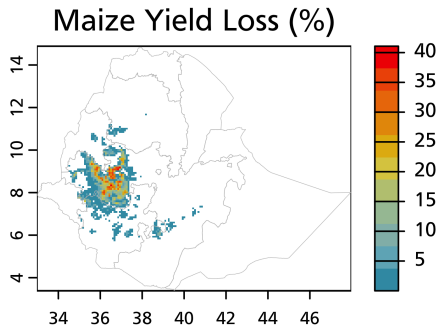
Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland



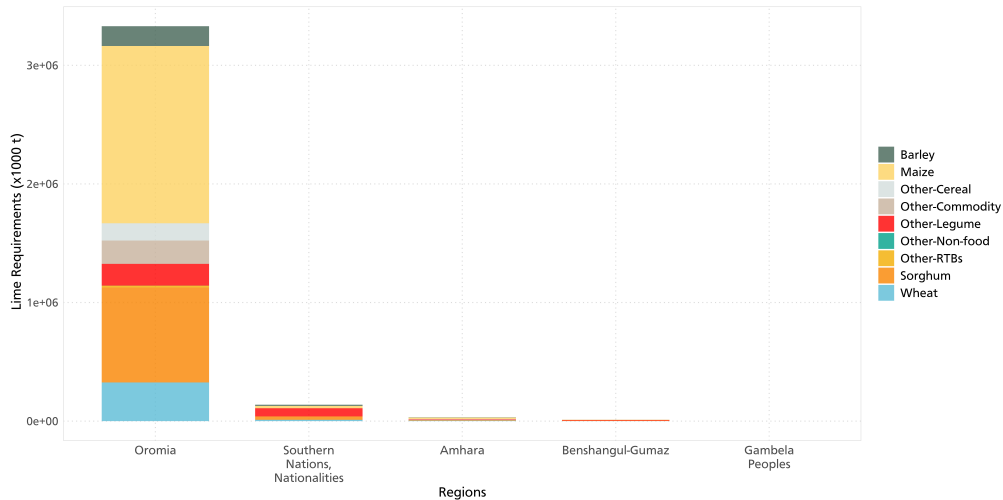
Total crop area in the top-10 most acidic Regions, not segregated by acidic cropland (table)

Regions	Barley	Maize	Other-Cereal	Other-Commodity	Other-Legume	Other-Non-food	Other-RTBs	Sorghum	Wheat
Amhara	359.74	426.54	199.97	8.31	233.79	32.16	26.94	608.07	499.49
Benshangul-Gumuz	1.18	45.14	32.09	2.72	17.73	0.47	1.31	67.40	3.23
Gambela Peoples	0.05	5.62	0.05	3.47	0.18	0.28	0.24	4.67	0.02
Oromia	484.59	1071.18	91.75	338.94	279.74	16.59	54.68	763.16	803.82
Southern Nations, Nationalities	92.72	277.53	5.50	117.43	88.85	54.53	38.53	118.07	139.81

Yield loss due to soil acidity for Maize and Sorghum in Ethiopia



Lime requirements (t) for acid soil remediation in Ethiopia

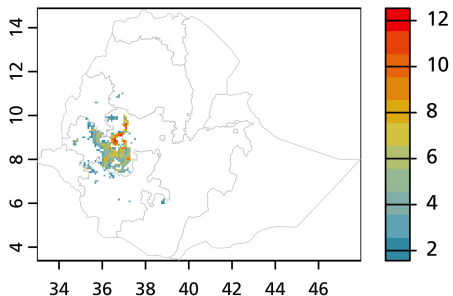


Lime requirements (1000 t) for acid soil remediation in Ethiopia (table)

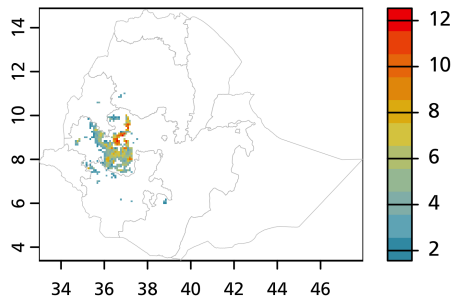
Regions	Barley	Maize	Other-Cereal	Other-Legume	Other-RTBs	Sorghum	Wheat	Other-Commodity	Other-Non-food
Amhara	2.79	11.61	3.71	5.11	0.88	3.93	4.39	-	-
Benshangul-Gumuz	0.06	1.70	2.08	5.43	0.12	3.12	0.88	-	-
Gambela Peoples	-	-	-	0.06	-	-	-	-	-
Oromia	166.72	1494.74	145.69	183.10	16.17	800.00	326.13	196.8	0.43
Southern Nations, Nationalities	9.90	19.54	0.35	68.67	1.43	27.58	10.79	-	-

Lime rate for Maize and Sorghum in Ethiopia

Maize Lime Rate (t/ha)

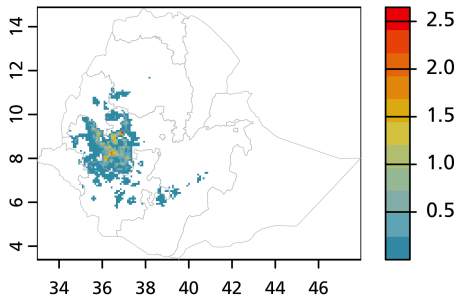


Sorghum Lime Rate (t/ha)

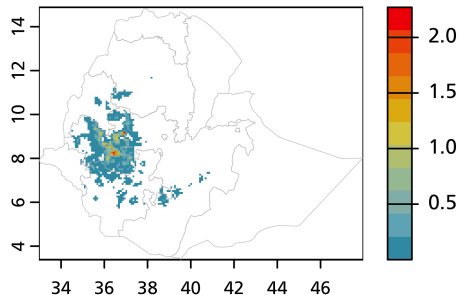


Yield response to lime application in Ethiopia

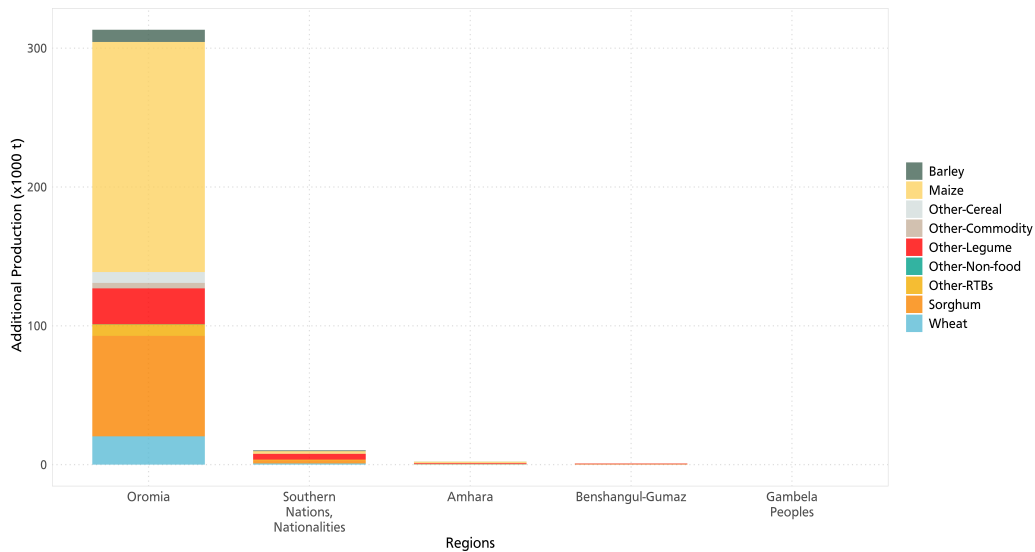
Maize Yield Response (t/ha)



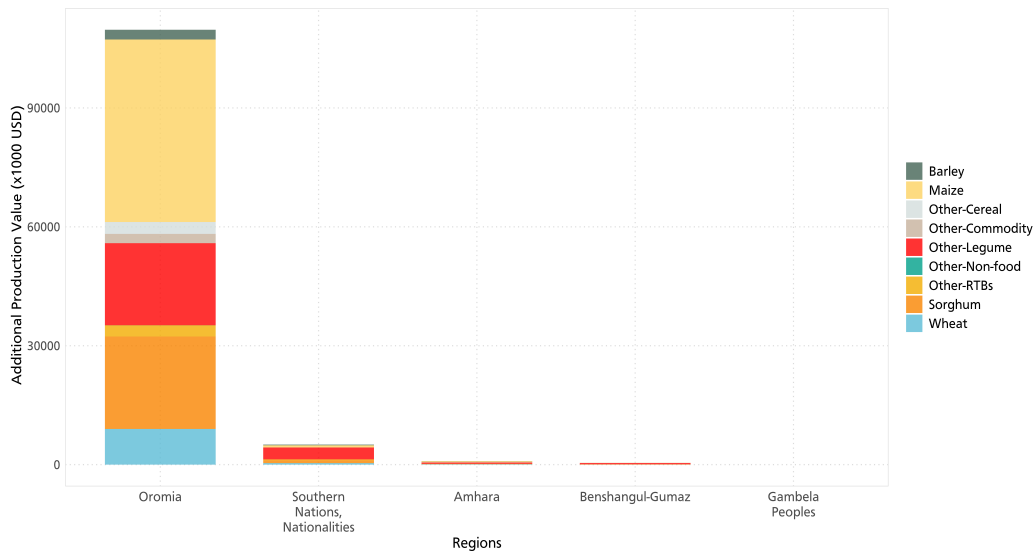
Sorghum Yield Response (t/ha)



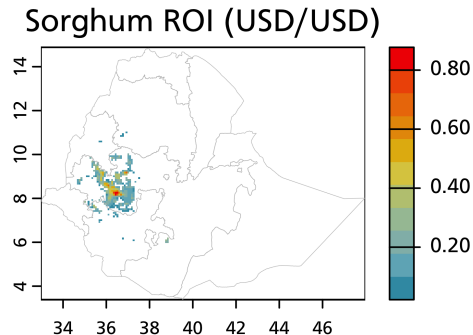
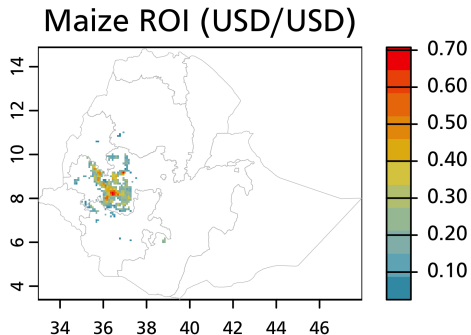
Potential for additional production (tone) in top 10 acidic Regions in Ethiopia



Potential for additional production value (US\$) in top 10 acidic Regions in Ethiopia

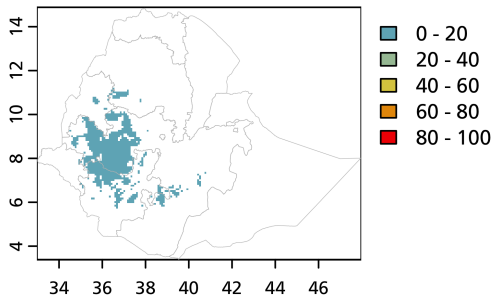


Return on investment for lime application in Ethiopia

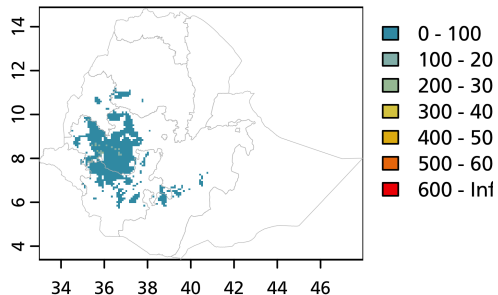


Profitability of lime application in Ethiopia

Profitable area (%)



Weighted profit (USD/ha)



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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